Eliminating finite algebraic choices with one coordinate

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Abstract

We give an explicit polynomial which replaces all finite algebraic choices in a triangular monic presentation by one rational coordinate. The proof includes degree drops, nonzero guards and inactive nonreduced fibers. We extend the construction to every number field using explicit anisotropic forms of degree e in e^2 variables. The results are unconditional algebraic statements; they require no arithmetic hypothesis about definability of the integers.

1 Finite choices and rational witnesses

An existential polynomial construction often contains coordinates which, once its other coordinates are fixed, have only finitely many possible values. We give an explicit elimination of all such finite choices using one further rational coordinate. The construction handles nonzero conditions, drops in degree and inactive nonreduced fibers. All products and determinants in the formula have explicit finite index sets.

The result is algebraic and does not depend on any assertion about Diophantine definitions of the integers. Its extension to arbitrary number fields uses an anisotropic form to combine its residuals.

2 An explicit polynomial for a guarded finite algebra

Let $R = \mathbb{Z}[x_1, \dots, x_b]$. Consider a triangular monic presentation

$$A = R[z_1, \dots, z_m]/(f_1, \dots, f_m), \qquad f_i \in R[z_1, \dots, z_i],$$

where f_i is monic of degree $e_i \geq 1$ in z_i. Its standard monomials $\prod_i z_i^{a_i}$, $0 \leq a_i < e_i$, form an R-basis. In particular A is free of rank $n = \prod_i e_i$. For $a \in A$, write M_a for its multiplication matrix in this ordered monomial basis. The matrices M_a commute, and their entries are integer polynomials in the base coordinates. They are completely specified by repeated reduction using the monic equations; they need not be materialized to specify a determinant involving them.

Fix elements $G, E_1, \dots, E_r \in A$. We assume that in every geometric fiber, every point at which G is nonzero has local length one. This condition holds, for example, if G contains the product of the diagonal derivatives $\partial f_i/\partial z_i$ as a factor. We seek the rational base points admitting a rational point of A with

$$G \neq 0, \qquad E_1 = \cdots = E_r = 0.$$

For $0 \leq \ell \leq mn(n+1)/2$, define

$$\lambda_\ell = 1 + \sum_{i=1}^{m} \ell^i z_i, \qquad M = M_{G\lambda_\ell}, \qquad H(V) = \det(I - VM) = \sum_{k=0}^{n} a_k V^k.$$

Here and below $a_k \in R$ depend on the fixed integer ℓ; always $a_0 = 1$. Also put

$$R_i(V) = \mathrm{Tr}\big(M_{E_i}\,\mathrm{adj}(I - VM)\big).$$

This is a polynomial in $R[V]$ of degree at most $n-1$.

For each $1 \le d \le n$, set

$$\begin{aligned} H_d(V) &= \sum_{k=0}^{d} a_k V^k, \\ g &= a_d\,\mathrm{Res}_V(H_d, H_d'), \qquad L = a_d^2 g, \\ \Psi(U) &= 1 + \sum_{j=1}^{d} a_{d-j} a_d^{2j-1} g^j U^j, \\ \widetilde{R}_i(U) &= (LU)^{n-1} R_i(1/(LU)), \\ Q_{\ell,d}(\mathbf{x}, U) &= \Psi(U)^2 + \sum_{k=d+1}^{n} a_k^2 + \sum_{i=1}^{r} \widetilde{R}_i(U)^2. \end{aligned}$$

The resultant is the Sylvester determinant with the specified degree bounds d and $d-1$. The expression for $\widetilde{R}_i$ denotes polynomial coefficient reversal, hence defines a polynomial also at $LU = 0$. Every displayed expression has integer coefficients.

Theorem 2.1. *The explicitly specified polynomial*

$$P(\mathbf{x}, U) = \prod_{\ell=0}^{mn(n+1)/2} \prod_{d=1}^{n} Q_{\ell,d}(\mathbf{x}, U)$$

has the following property. For every $\mathbf{a} \in \mathbb{Q}^b$, there exists $U \in \mathbb{Q}$ with $P(\mathbf{a}, U) = 0$ if and only if there is a rational point of the fiber algebra $A_{\mathbf{a}}$ satisfying $G \neq 0$ and all $E_i = 0$. The formula uses only one coordinate in addition to the b base coordinates; its formal algebra generators are eliminated by the determinants and are not quantified coordinates.

Proof. Fix a rational base point. First consider one factor $Q_{\ell,d}$. At $a_d g = 0$ or $U = 0$, its reciprocal residual Ψ equals one. Otherwise, with $V = 1/(LU)$,

$$\Psi(U) = (LU)^d \frac{H_d(V)}{a_d}.$$

Thus a rational zero of $Q_{\ell,d}$ implies $a_d g U \neq 0$, all coefficients a_k with $k > d$ vanish, and $H(V) = H_d(V) = 0$. The resultant condition makes this a simple root of H. Since $H(0) = 1$, the root V is nonzero.

The rational matrix $I - VM$ has a one-dimensional kernel. Every multiplication matrix M_{z_i} preserves it and therefore acts on it by a rational scalar ζ_i. Applying the algebra relations to this common eigenline shows that $(\zeta_1, \ldots, \zeta_m)$ is a rational point of the fiber. Its $G\lambda_\ell$-value is $1/V$, so its G-value is nonzero.

To verify the closed conditions, write $J = \mathrm{Tr}(\mathrm{adj}(I - VM))$. At a root of H, the adjugate identity gives $M\,\mathrm{adj}(I - VM) = V^{-1}\,\mathrm{adj}(I - VM)$. Differentiating the determinant then yields

$$J = -VH'(V) \neq 0.$$

The columns of the adjugate lie in the common eigenline, so

$$R_i(V) = E_i(\boldsymbol{\zeta}) J.$$

Since $LU \neq 0$, the equation $\widetilde{R}_i(U) = 0$ is equivalent to $R_i(V) = 0$, and hence $E_i(\boldsymbol{\zeta}) = 0$. This proves the reverse implication without assuming that the rest of the fiber algebra is reduced.

Conversely, suppose a suitable rational point exists. Over an algebraic closure, multiplication by $G\lambda_\ell$ has the value $G(p)\lambda_\ell(p)$ at every geometric point p, with multiplicity equal to its local length. If $G(p) = 0$, its factor in $\det(I - VM)$ is one, even if the local algebra is nonreduced. If $G(p) \neq 0$, its local length is one by hypothesis.

At any such point p, the polynomial $G(p)(1 + \sum_{i=1}^{m} \ell^i z_i(p))$ is nonzero as a polynomial in ℓ, because its constant coefficient is $G(p) \neq 0$. For distinct such points p, q, their difference is also nonzero: if all coefficients vanished, first $G(p) = G(q) \neq 0$, and then all their coordinates would agree. Each of these polynomials has degree at most m. With at most n geometric points, there are at most $mn(n+1)/2$ bad values of ℓ in total, counting zero values and collisions. Thus one of the specified integers makes all nonzero guarded eigenvalues distinct and nonzero.

For that ℓ, the polynomial H is squarefree of some degree $1 \leq d \leq n$. The given rational point provides a nonzero rational root V of H. Its closed residuals vanish, so the trace identities give $R_i(V) = 0$. The leading coefficient and resultant are nonzero. Taking $U = 1/(LV)$ therefore gives a zero of the corresponding factor $Q_{\ell,d}$, and hence of P. □

The degree cases are part of the literal finite formula. They include every exceptional drop in degree of H. The discriminant condition excludes bad separating choices, and the finite bound on ℓ proves that this exclusion loses no valid rational point. In particular the construction does not leave a generic primitive-element argument or an unperformed exceptional-stratum decomposition implicit.

3 Extension to arbitrary number fields

Definition 3.1. A homogeneous polynomial $A \in F[X_1, \ldots, X_s]$ is anisotropic over F if its only zero in F^s is the zero tuple.

Lemma 3.2. *For every number field F and every integer $e \geq 1$ there is an explicit anisotropic form of degree e in e^2 variables over F.*

Proof. Choose a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_F$, an element $\pi \in \mathfrak{p} \setminus \mathfrak{p}^2$, and a monic polynomial $g \in \mathcal{O}_F[Z]$ of degree e whose reduction is irreducible over the residue field. Such a polynomial is obtained by lifting the coefficients of any irreducible degree-e polynomial over that finite field. Let $N_g(\mathbf{X})$ be the determinant of multiplication by $\sum_{i=0}^{e-1} X_i Z^i$ on $F[Z]/(g)$.

For a nonzero tuple $\mathbf{x} \in F^e$, put $r = \min_i v_{\mathfrak{p}}(x_i)$. After dividing its coordinates by π^r, their reductions are not all zero. Multiplication by their reduced polynomial is invertible in the residue field extension defined by g. Its determinant is therefore a unit. By homogeneity,

$$v_{\mathfrak{p}}(N_g(\mathbf{x})) = er.$$

Consequently

$$A_e(\mathbf{X}_0, \ldots, \mathbf{X}_{e-1}) = \sum_{j=0}^{e-1} \pi^j N_g(\mathbf{X}_j)$$

is anisotropic: the nonzero summands have pairwise distinct valuations modulo e, so the summand of least valuation cannot cancel. This is a homogeneous degree-e form in e^2 coordinates. □

Theorem 3.3. *Replace $\mathbb{Z}$ by $\mathcal{O}_F$ and $\mathbb{Q}$ by a number field F in the finite algebra presentation of Section 2. Retain the same rank n, guard G and r closed residuals. There is an explicitly specified polynomial in the same base coordinates and one additional coordinate whose F-solutions have exactly the required input projection.*

Proof. All multiplication matrices, determinants, resultants, reciprocal polynomials and trace residuals in Theorem 2.1 are defined over $\mathcal{O}_F$ by the same formulas. For fixed ℓ, d, replace the sum of squares defining $Q_{\ell,d}$ by an anisotropic form applied to the list

$$\Psi(U), \quad a_{d+1}, \dots, a_n, \quad \widetilde{R}_1(U), \dots, \widetilde{R}_r(U).$$

The list has $1 + n - d + r \le n + r$ entries. Take $e = \lceil \sqrt{n+r} \rceil$, use the form of Lemma 3.2, and pad the list with zeros. Its vanishing is equivalent to the vanishing of every listed entry.

The proof of the finite algebra theorem now applies verbatim: it uses characteristic zero, the finite bound on bad separating integers, and the one-dimensional kernel at a simple rational root. The sole use of the ordering of $\mathbb{Q}$ was to combine the listed residuals, and that step has been replaced by anisotropy. Multiplication over all the finite choices gives the required polynomial. Its coefficients belong to $\mathcal{O}_F$. □

For each factor, if the largest degree of the listed residuals is $D_{\ell,d}$, its replacement has degree at most $\lceil \sqrt{n+r} \rceil D_{\ell,d}$. Over a number field with a real embedding, sums of squares give instead the bound $2D_{\ell,d}$. Both methods preserve the number of coordinates. The combining form depends on F; this theorem does not assert that the original rational polynomial has the same projection after arbitrary base change.